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Uncovering ChatGPT's Capabilities in Recommender Systems

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ABSTRACT

The debut of ChatGPT has recently attracted significant attention from the natural language processing (NLP) community and beyond. Existing studies have demonstrated that ChatGPT shows significant improvement in a range of downstream NLP tasks, but the capabilities and limitations of ChatGPT in terms of recommendations remain unclear. In this study, we aim to enhance ChatGPT's recommendation capabilities by aligning it with traditional information retrieval (IR) ranking capabilities, including point-wise, pair-wise, and list-wise ranking. To achieve this goal, we re-formulate the aforementioned three recommendation policies into prompt formats tailored specifically to the domain at hand. Through extensive experiments on four datasets from different domains, we analyze the distinctions among the three recommendation policies. Our findings indicate that ChatGPT achieves an optimal balance between cost and performance when equipped with list-wise ranking. This research sheds light on a promising direction for aligning ChatGPT with recommendation tasks. To facilitate further explorations in this area, the full code and detailed original results are open-sourced at https://github.com/rainym00d/LLM4RS.

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CCS CONCEPTS

• **Information systems** → **Recommender systems**.

KEYWORDS

ChatGPT, large language model, recommender systems

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1 INTRODUCTION

Large language models (LLMs), such as ChatGPT developed by OpenAI [23], have recently gained significant attention from the natural language processing (NLP) community and beyond. These LLMs possess a versatile nature and extensive world knowledge, allowing them to be applied not only in various NLP tasks [3, 5, 16], but also in domains such as education [11, 19, 22], medicine [1, 2, 29], search [20, 26, 37] and law [4, 8, 38].

Meanwhile, previous research has indicated that off-the-shelf pre-trained language models (LMs) can be directly used as recommenders by adapting recommendation tasks into multi-token cloze tasks using prompts [25, 34, 43]. Hence, a natural research question arises regarding how to effectively align LLMs with recommendation capabilities and whether LLMs can work as few-shot or zero-shot recommender systems.

The primary objective of recommender systems is to alleviate information overload by providing personalized top-K item ranking lists for users [32]. In information retrieval (IR), previous studies